



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004

**HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 1**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in **3** sections.
Section A (Questions 1 - 5), Section B (Questions 6 - 9) and Section C (Questions 10 - 13).
- Start each **NEW** section in a separate answer booklet.

Total Marks - 90 Marks

- Attempt Sections A - C
- All questions are **NOT** of equal value.

Examiner: *E. Choy*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 90
Attempt Questions 1 – 13
All questions are NOT of equal value

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

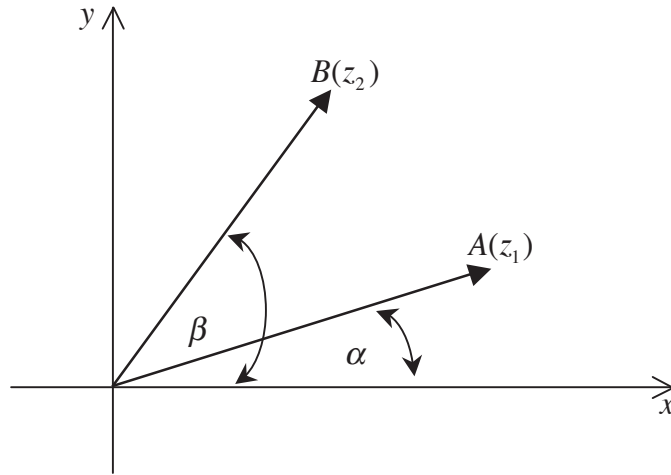
SECTION A (Use a SEPARATE writing booklet)

		Marks
Question 1 (8 marks)		
(a)	Evaluate $(3 + 4i) \div (1 + i)$	2
(b)	If $(x + iy)(2 + 3i) = 5 + 6i$ find x and y	2
(c)		
(i)	Express $\frac{-1 - i\sqrt{3}}{1 - i}$ in modulus-argument form	2
(ii)	Hence evaluate $\left(\frac{-1 - i\sqrt{3}}{1 - i}\right)^6$	2
Question 2 (8 marks)		
If P represents the complex number z , sketch the locus of P (on separate diagrams) if:		
(i)	$ z - 1 = 4$	2
(ii)	$-1 \leq \text{Im}(z) \leq 2$	2
(iii)	$-\frac{\pi}{4} \leq \arg(z) \leq \frac{2\pi}{3}$	2
(iv)	$\arg\left(\frac{z - i}{z - 1}\right) = \frac{\pi}{6}$	2
Question 3 (6 marks)		
(a)	If z is a non zero complex number such that $z + 1/z$ is real, prove that $\text{Im}(z) = 0$ or $ z = 1$	3
(b)	Find the square roots of $-2 - 2i$.	3
Leave your answer in modulus-argument form.		

SECTION A (continued)

Question 4 (4 marks)

Marks



In the diagram $\arg(z_1) = \alpha$ and $\arg(z_2) = \beta$.

4

If $|z_1| = |z_2|$ prove that $\arg(z_1 z_2) = \arg((z_1 + z_2)^2)$

Question 5 (4 marks)

The point A in an Argand diagram represents the complex number $3 + 4i$.

4

Find the complex number represented by B if $\triangle OAB$ is an equilateral triangle with B in the fourth quadrant.

O represents the complex number 0 .

Leave your answer in the form $a + ib$.

SECTION B (Use a SEPARATE writing booklet)

Question 6 (7 marks)

Marks

Given $P(x) = x^4 - 2x^3 + 6x^2 - 2x + 5$

- | | | |
|------|---|---|
| (i) | Find the zeros of $P(x)$ given that $1 + 2i$ is a zero. | 3 |
| (ii) | Express $P(x)$ in factored form: | |
| (α) | over the complex field; | 2 |
| (β) | over the real field. | 2 |

Question 7 (9 marks)

If α, β and γ are the roots of the equation $x^3 - 2x + 5 = 0$, find the equation which has roots:

- | | | |
|-------|---|---|
| (i) | $2/\alpha, 2/\beta, 2/\gamma$; | 3 |
| (ii) | $\alpha^2, \beta^2, \gamma^2$; | 3 |
| (iii) | $\alpha + \beta, \beta + \gamma, \gamma + \alpha$ | 3 |

SECTION B (continued)

Question 8 (6 marks)

Marks

(a) Show that $f(x) = x^n - 1$ has no multiple roots, where n is an integer with $n > 1$.

3

(b) If the roots of $x^n - 1 = 0$ are $1, \alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{n-1}$ show that

3

$$(1 - \alpha_1)(1 - \alpha_2)(1 - \alpha_3) \cdots (1 - \alpha_{n-1}) = n$$

Question 9 (9 marks)

Consider $\omega = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$

(i) Prove that $\omega^5 = 1$ and $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$

3

(ii) Prove that $z = \omega + 1/\omega$ is a root of $z^2 + z - 1 = 0$

3

(iii) Hence prove that $\cos \frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}$

3

SECTION C (Use a SEPARATE writing booklet)

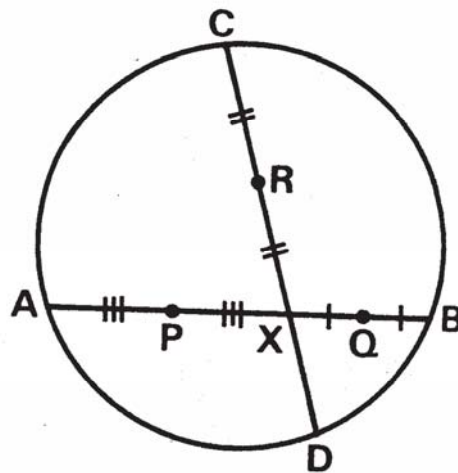
Question 10 (6 marks)

Marks

Given that $y = x^3 - 3px + q$ where $p, q \in \mathbb{R}$

- | | | |
|------|--|---|
| (i) | Find the coordinates of the stationary points (in terms of p and q) of $y = f(x)$. | 3 |
| (ii) | Hence, find the relationship between p and q for $f(x) = x^3 - 3px + q$ to have 3 distinct real roots. | 3 |

Question 11 (6 marks)



6

AB and CD are two chords of a circle intersecting at a point X .
 P , Q and R are the midpoints of AX , XB and CX respectively.

Prove that the circle PQR also bisects DX .

SECTION C (continued)

Question 12 (11 marks)

A and B are two points in an Argand diagram presenting the complex numbers $z_1 = -1$ and $z_2 = \cos \theta + i \sin \theta$ respectively,

where $\frac{\pi}{2} < \theta < \pi$.

C is the point representing the complex number $z_3 = z_1 + z_2$.

- (i) Sketch the quadrilateral $OACB$ in an Argand diagram, where O is the point representing the complex number 0. 3

Mark an angle in the diagram which is equal to θ .

- (ii) Let $z_4 = z_2 - z_1$

(α) Show that $\frac{z_4}{z_3} = i \left(\frac{\sin \theta}{\cos \theta - 1} \right)$. Hence find $\arg \left(\frac{z_4}{z_3} \right)$. 4

(β) Using (α) show that the diagonals of the quadrilateral $OACB$ are perpendicular to each other. 4

Question 13 (6 marks)

How many ways are there to place nine different rings on the four fingers of your right hand (excluding the thumb) if:

- (i) the order of the rings on a finger does not matter? 3
- (ii) the order of the rings on a finger is considered? 3

THIS IS THE END OF THE PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$



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Mathematics Extension 2

Sample Solutions

$$1(a) \frac{3+4i}{1+i} \times \frac{1-i}{1-i} = \frac{3-3i+4i+4}{1+1}$$

$$= \frac{7}{2} + \frac{i}{2} \quad \checkmark$$

(b) Method 1: $(x+iy)(2+3i) = 2x+3ix+2iy-3y$

Equating real & imaginary parts,

$$2x - 3y = 5 \quad \text{--- (1)}$$

$$3x + 2y = 6 \quad \text{--- (2)}$$

$$\textcircled{1} \times 2: \quad 4x - 6y = 10 \quad \text{--- (3)}$$

$$\textcircled{2} \times 3: \quad 9x + 6y = 18 \quad \text{--- (4)}$$

$$\textcircled{3} + \textcircled{4}: \quad 13x = 28$$

$$x = 28/13 \quad \checkmark$$

$$2y = 6 - 3(28/13)$$

$$y = -3/13 \quad \checkmark$$

Method 2: $x+iy = \frac{5+6i}{2+3i} \times \frac{2-3i}{2-3i}$

$$= \frac{10-15i+12i+18}{4+9}$$

$$= \frac{28-3i}{13}$$

$$\therefore x = \frac{28}{13}, \quad y = -\frac{3}{13}$$

$$1(c) (i) \quad -1 - i\sqrt{3} = 2 \operatorname{cis}(-2\pi/3) \quad \checkmark$$

$$1 - i = \sqrt{2} \operatorname{cis}(-\pi/4) \quad \checkmark$$

$$\therefore \frac{-1 - i\sqrt{3}}{1 - i} = \sqrt{2} \operatorname{cis}\left(-\frac{5\pi}{12}\right) \quad \checkmark$$

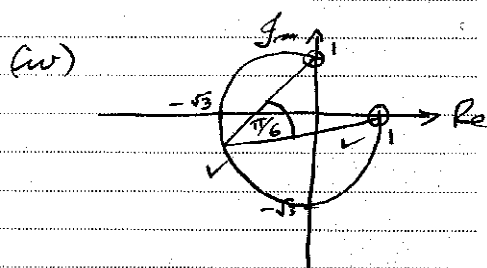
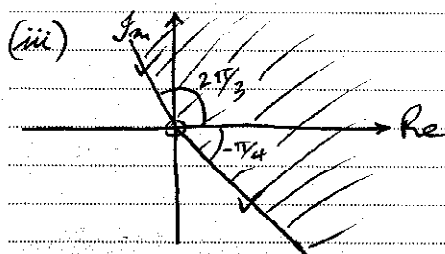
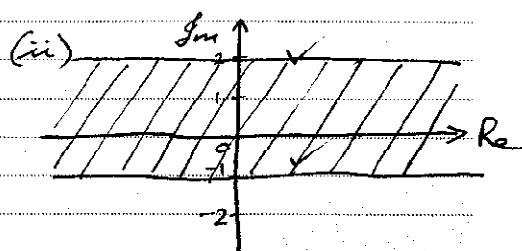
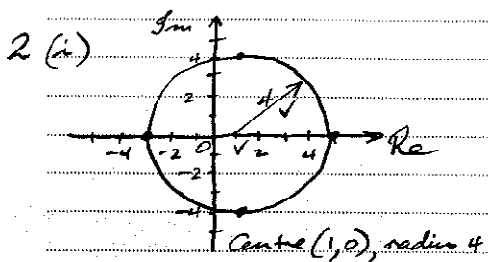
$$\frac{-2}{3} + \frac{1}{4} = \frac{-8+3}{12}$$

$$\frac{1}{2} \text{ for } \frac{-1+\sqrt{3}}{2} - i\left(\frac{1+\sqrt{3}}{2}\right)$$

$$(ii) \quad \left(\frac{-1 - i\sqrt{3}}{1 - i}\right)^6 = 8 \operatorname{cis}\left(-\frac{30\pi}{12}\right) \quad \checkmark$$

$$= 8 \operatorname{cis}(-5\pi/2) \quad \checkmark$$

$$= -8i \quad \checkmark$$



3(a) Let $z = a + ib$, then

$$z + \frac{1}{z} = a + ib + \frac{1}{a + ib} \times \frac{a - ib}{a - ib}$$

$$= a + ib + \frac{a - ib}{a^2 + b^2}$$

As $z + \frac{1}{z}$ is real, $b - \frac{b}{a^2 + b^2} = 0 \quad \checkmark$

ie. $b\left(1 - \frac{1}{a^2 + b^2}\right) = 0$

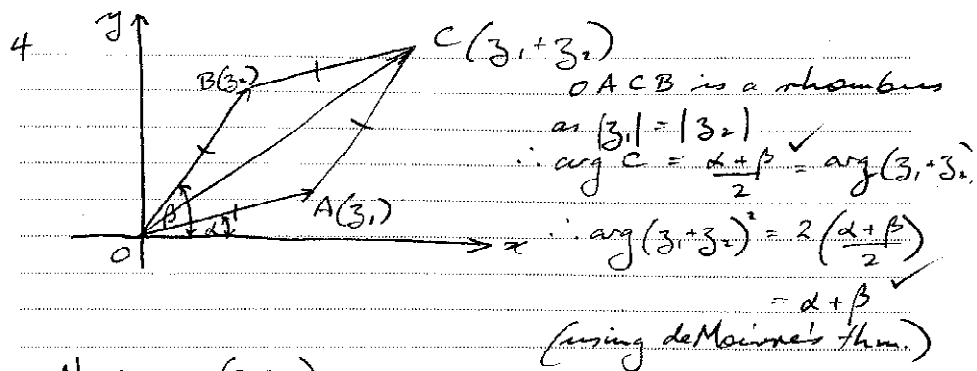
so $b = 0$ or $a^2 + b^2 = 1 \quad \checkmark$

$\therefore \operatorname{Im}(z) = 0$ or $|z| = 1 \quad \checkmark$

(b) $-2 - 2i = \sqrt{8} \operatorname{cis}\left(-3\pi/4 + 2n\pi\right) \quad \checkmark$

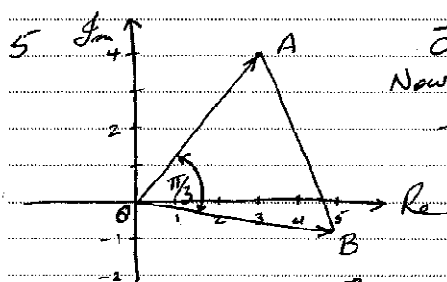
$\therefore \sqrt{-2 - 2i} = 8^{1/4} \operatorname{cis}\left(-3\pi/8 + n\pi\right)$

$$= 8^{1/4} \operatorname{cis}\left(-3\pi/8\right), 8^{1/4} \operatorname{cis}\left(5\pi/8\right) \quad \checkmark$$



Now, $\arg(z_1 z_2) = \arg z_1 + \arg z_2$
 $= \alpha + \beta$

$\therefore \arg(z_1 z_2) = \arg(z_1 + z_2)^2$



$\vec{OA} = 3 + 4i$
 Now, unit vector with $\arg -\pi/3$
 is $\cos(-\pi/3) = \frac{1}{2} - \frac{i\sqrt{3}}{2}$

$\vec{OB} = \vec{OA} \times \cos(-\pi/3)$
 $= (3 + 4i) \times \left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)$

$= \frac{1}{2}(3 - 3i\sqrt{3} + 4i + 4\sqrt{3})$

$= \frac{3 + 4\sqrt{3}}{2} + \frac{i}{2}(4 - 3\sqrt{3})$

$\approx 4.9641 - 0.5981i$

Question 6

$$P(x) = x^4 - 2x^3 + 6x^2 - 2x + 5$$

- (i) Since $x = 1 + 2i$ is a zero **AND** the coefficients are real then $x = 1 - 2i$ is also a zero (*conjugate root theorem*).

So $x - (1 + 2i) \quad x - (1 - 2i) = x^2 - 2x + 5$ is a factor of $P(x)$.

$$\text{So } P(x) = x^4 - 2x^3 + 6x^2 - 2x + 5 = (x^2 - 2x + 5)(x^2 + bx + 1)$$

Collecting terms of degree 1 and comparing coefficients we get
 $-2 + 5b = -2 \quad b = 0$

[Or by applying long division methods ie

$$(x^4 - 2x^3 + 6x^2 - 2x + 5) \div (x^2 - 2x + 5) \text{ etc....}]$$

Thus $P(x) = x^4 - 2x^3 + 6x^2 - 2x + 5 = (x^2 - 2x + 5)(x^2 + 1)$, so the zeros of $P(x)$ are $x = 1 + 2i, 1 - 2i, \pm i$

- (ii) (α)

$$P(x) = x^4 - 2x^3 + 6x^2 - 2x + 5 = (x - (1 + 2i))(x - (1 - 2i))(x - i)(x + i)$$

$$(\beta) \quad P(x) = x^4 - 2x^3 + 6x^2 - 2x + 5 = (x^2 - 2x + 5)(x^2 + 1)$$

Question 7

$$x^3 - 2x + 5 = 0$$

- (i) Apply the transformation $y = \frac{2}{x} \quad x = \frac{2}{y}$

$$\text{So } \frac{2}{y}^3 - 2\frac{2}{y} + 5 = 0 \quad \frac{8}{y^3} - \frac{4}{y} + 5 = 0 \quad 5y^3 - 4y^2 + 8 = 0$$

So the equation with roots $2/\alpha, 2/\beta, 2/\gamma$ is $5x^3 - 4x^2 + 8 = 0$

- (ii) Apply the transformation $y = x^2$

$$\text{NB } x^3 - 2x + 5 = 0 \quad x(x^2 - 2) = -5$$

So square both sides and we get $x^2(x^2 - 2)^2 = 25 \quad y(y - 2)^2 = 25$

So the equation with roots $\alpha^2, \beta^2, \gamma^2$ is given by

$$x \left((x - 2)^2 = 25 \text{ OR } x^3 - 4x^2 + 4x - 25 = 0 \right)$$

- (iii) We have to use the fact that $\alpha + \beta + \gamma = 0$ $\alpha + \beta = -\gamma$ and so on. So use the transformation $y = -x$ $x = -y$

$$\text{So we get } (-y)^3 - 2(-y) + 5 = 0 \quad -y^3 + 2y + 5 = 0.$$

$$\text{So the equation with roots } \alpha + \beta, \beta + \gamma, \gamma + \alpha \text{ is } \boxed{x^3 - 2x - 5 = 0}$$

Question 8

(a) $f(x) = x^n - 1$ $f'(x) = nx^{n-1}$

$$\text{Suppose } \alpha \text{ is a multiple root then } f(\alpha) = f'(\alpha) = 0 \quad n\alpha^{n-1} = 0 \quad \alpha = 0$$

BUT $f(0) \neq 0$, so $f(x)$ cannot have any multiple roots.

(b) $x^n - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1})$

$$\text{By long division; series; or other means } x^n - 1 = (x - 1)(1 + x + x^2 + \dots + x^{n-1})$$

$$\text{So } x^n - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1}) = (x - 1)(1 + x + x^2 + \dots + x^{n-1})$$

$$\text{So } (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1}) = 1 + x + x^2 + \dots + x^{n-1}, \text{ sub } x = 1 \text{ into both}$$

sides

$$(1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = (1 + 1 + 1^2 + \dots + 1^{n-1}) = n \quad \mathbf{QED}$$

Question 9

$$\omega = \text{cis} \frac{2\pi}{5} \neq 1$$

(i) $\omega^5 = \text{cis} \ 5 \times \frac{2\pi}{5} = \text{cis} 2\pi = 1. \text{ [de Moivre's Theorem - DMT]}$

$$\omega^5 - 1 = (\omega - 1)(1 + \omega + \omega^2 + \omega^3 + \omega^4) = 0$$

$$\because \omega \neq 1 \quad 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$$

$$\mathbf{OR} \ 1 + \omega + \omega^2 + \omega^3 + \omega^4 = \frac{\omega^5 - 1}{\omega - 1} = \frac{1 - 1}{\omega - 1} = 0 \text{ [using geometric series]}$$

(ii) $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0 \quad \frac{1}{\omega^2} + \frac{1}{\omega} + 1 + \omega + \omega^2 = 0 \text{ [} \div \omega^2 \text{]}$

$$\text{Let } z = \omega + \frac{1}{\omega} \quad (z^2 - 2) + 1 + z = 0 \quad z^2 + z - 1 = 0$$

$$[\because \frac{1}{\omega^2} + \omega^2 = \omega + \frac{1}{\omega}^2 - 2]$$

(ii) **Alternative solution**

Examine $z^2 + z - 1$ when $z = \omega + \frac{1}{\omega}$

$$\begin{aligned} & z^2 + z - 1 \\ &= \left(\omega + \frac{1}{\omega}\right)^2 + \omega + \frac{1}{\omega} - 1 \\ &= \omega^2 + 2 + \frac{1}{\omega^2} + \omega + \frac{1}{\omega} - 1 \\ &= \frac{\omega^4 + \omega^3 + \omega^2 + \omega + 1}{\omega^2} \\ &= \frac{0}{\omega^2} \\ &= 0 \end{aligned}$$

So $z = \omega + \frac{1}{\omega}$ is a solution of $z^2 + z - 1 = 0$

(iii) $|\omega| = 1 \quad \frac{1}{\omega} = \bar{\omega} \quad \omega + \frac{1}{\omega} = 2 \operatorname{Re} \omega = 2 \cos \frac{2\pi}{5}$

$$z^2 + z - 1 = 0 \quad z = \frac{-1 \pm \sqrt{5}}{2}$$

$$\because \cos \frac{2\pi}{5} > 0 \quad 2 \cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{2}$$

$$\cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$$

[NB the other solution of the quadratic is $2 \cos \frac{4\pi}{5}$]

SECTION C

Q(10) $y = x^3 - 3px + q$

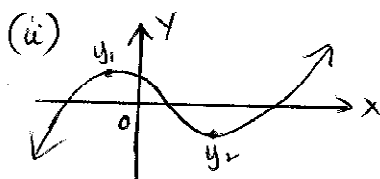
(i) $y' = 3x^2 - 3p = 0 \therefore$ st. pts

when $x = \pm\sqrt{p}$

When $x = \sqrt{p}$, $y = -2p^{3/2} + q$

When $x = -\sqrt{p}$, $y = 2p^{3/2} + q$

$\therefore$ St. pts $(\sqrt{p}, q - 2p^{3/2})$
 $(-\sqrt{p}, q + 2p^{3/2})$



For 3 distinct real roots

$y_1, y_2 < 0 \Rightarrow (q - 2p^{3/2})(q + 2p^{3/2}) < 0$

$\therefore q^2 < 4p^3$

or $|q| < 2p\sqrt{p}$

Q(11) Let E be the point where circle PQR meets DX.

Now $CX \cdot DX = AX \cdot BX$

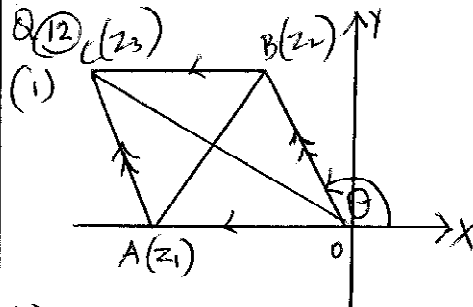
(product of chords of circle)

$\therefore (2RX) \cdot DX = (2PX)(2QX)$

$\Rightarrow (RX) \cdot \frac{1}{2}(DX) = (PX)(QX)$

However $(RX)(EX) = (PX)(QX)$

$(EX) = \frac{1}{2}(DX)$



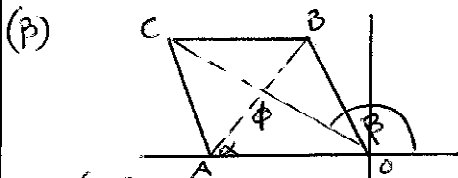
(i) $z_4 = z_2 - z_1$
 $= (\cos\theta + i) + i\sin\theta$

(ii) $z_3 = z_1 + z_2 = (-1 + \cos\theta) + i\sin\theta$

$\therefore \frac{z_4}{z_3} = \frac{(\cos\theta + 1) + i\sin\theta}{(-1 + \cos\theta) + i\sin\theta} \cdot \frac{(\cos\theta - 1) - i\sin\theta}{(\cos\theta - 1) - i\sin\theta}$
 $= \frac{-2i\sin\theta}{2(1 - \cos\theta)}$ on simplifying
 $= -i \left[\frac{\sin\theta}{1 - \cos\theta} \right]$

and $\arg\left(\frac{z_4}{z_3}\right) = -\frac{\pi}{2}$ since

$\frac{\sin\theta}{1 - \cos\theta} < 0$ for $\frac{\pi}{2} < \theta < \pi$



$\arg\left(\frac{z_4}{z_3}\right) = \arg z_4 - \arg z_3 = -\frac{\pi}{2}$
 $\Rightarrow \arg z_3 - \arg z_4 = \frac{\pi}{2}$

$\arg z_3 = \beta$, $\arg z_4 = \alpha$

$\arg z_3 - \arg z_4 = \beta - \alpha = \phi$

$\therefore \phi = \frac{\pi}{2}$

Diagonals are $\perp$

or proved apl. using $\times$ by i

Q(13) (i) 4^9 (ii) $\frac{12!}{3! \times 9!} \times 9! = \frac{12!}{3!}$